Resampling Methods

Cross-Validation & Bootstrap — ISLR Chapter 5

Machine Learning with R

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Week 3, 2025

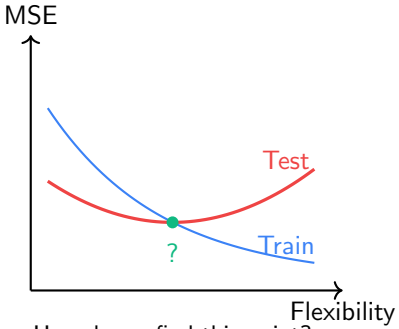
The Central Question

Weeks 1–2: We fit models
Today: How good are they *really*?

Two problems:

1. **Model assessment:**
   How well does the model predict new data?
2. **Model selection:**
   Which model (or tuning parameter) is best?

Both require estimating **test error**



How do we find this point?

Why Not Use Training Error?

Training MSE $= \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$

Problem:

Training MSE **always** decreases with flexibility

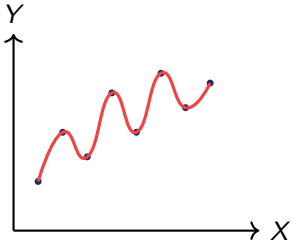
It can even reach **zero** (overfitting!)

Low training error $\neq$ good prediction

What we need:

Test error on **new, unseen data**

But we usually don't have a test set!



Train MSE = 0!
But terrible on new data

Solution: Resampling

Today's Roadmap

Cross-Validation:

1. Validation Set Approach
2. Leave-One-Out CV (LOOCV)
3. k-Fold CV
4. Bias-Variance trade-off for k

Bootstrap:

5. The bootstrap idea
6. Estimating uncertainty
7. Bootstrap for regression

Datasets:

`Auto`, `Default`, `Portfolio`

Approach 1: The Validation Set

Simplest idea:

1. Randomly split data into two halves
2. Train on one half
3. Evaluate on the other half

Example: Auto data, $n = 392$
Split: 196 train, 196 test
Fit: `mpg` $\sim$ `poly(horsepower, d)`
Compute test MSE for $d = 1, 2, \ldots, 10$

Train	Validation
196 obs	196 obs

Problems:

- High **variance**: different splits $\Rightarrow$ different MSEs
- Only half the data for training $\Rightarrow$ **overestimates** test error

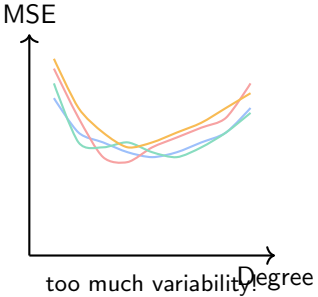
Validation Set: The Variance Problem

10 different random splits:
Each gives a different test MSE curve!

The **minimum** varies: sometimes $d = 2$, sometimes $d = 5$

This is **unreliable** for model selection

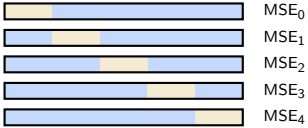
Root cause:
With only $n/2$ training points, the model has high variance
Results are too dependent on which observations end up in which half



Solution: Use more of the data

Approach 2: Leave-One-Out CV (LOOCV)

Idea: Train on $n-1$ observations, test on the 1 left out. Repeat n times.



$\vdots$ (repeat n times)

$$\text{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\text{MSE}_i = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i^{(-i)})^2$$

where $\hat{y}_i^{(-i)}$ = prediction for i from model trained **without** observation i

LOOCV: Pros, Cons, and a Shortcut

Advantages:

- Nearly unbiased (trains on $n-1$)
- No randomness in the split
- Uses almost all data for training

Disadvantages:

- Expensive: fit model n times
- High variance (correlated estimates)

Magic shortcut (linear models only!):

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1-h_i}\right)^2$$

where h_i = leverage of observation i

Requires fitting the model only **once**!

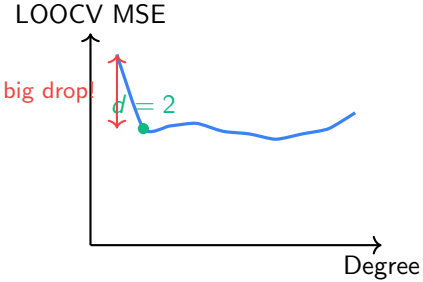
Inflates residuals for high-leverage points by exactly the right amount

LOOCV: Worked Example on Auto

Fit: $\texttt{mpg} \sim \texttt{poly}(\texttt{horsepower}, d)$ for $d = 1, \ldots, 10$

Degree d	LOOCV MSE
1	24.23
2	19.25
3	19.33
4	19.42
5	19.03

Sharp drop from linear to quadratic!
No clear gain beyond $d = 2$



R: `cv.glm(Auto, glm.fit)$delta[1]`

Approach 3: k-Fold Cross-Validation

Split data into k equal folds

1. Hold out fold 1 $\Rightarrow$ test; train on folds 2–k
2. Hold out fold 2 $\Rightarrow$ test; train on rest

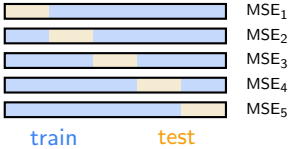
   $\vdots$

k. Hold out fold k $\Rightarrow$ test; train on rest

$$\mathrm{CV}_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{MSE}_i$$

Typical: $k = 5$ or $k = 10$
LOOCV = special case where $k = n$

5-fold CV:



Only fit model k times!
(vs. n times for LOOCV)

k-Fold CV: The Bias-Variance Trade-Off

Method	Bias	Variance	Computation
Validation set ($k = 2$)	High	High	Fast
5-fold CV	Medium	Low	Fast
10-fold CV	Low	Low	Medium
LOOCV ($k = n$)	Very low	High	Slow

Why does LOOCV have high variance?

We average n estimates, each trained on **nearly identical** data

$\Rightarrow$ Estimates are highly **correlated**

$\Rightarrow$ Averaging correlated quantities reduces variance less

Why does $k = 5$ have more bias?

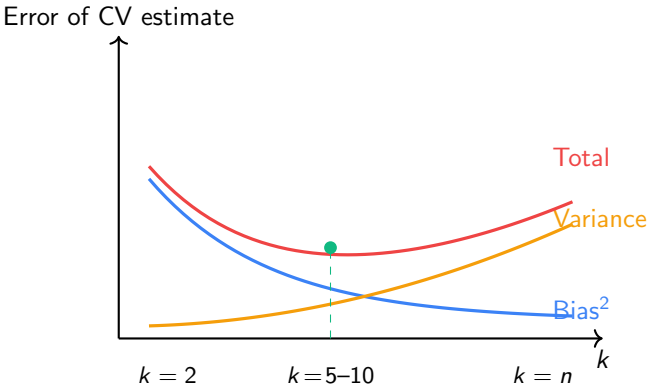
Each fold trains on $(k-1)/k$ of data

$k = 5$: trains on 80%

$k = n$: trains on $\approx 100\%$

Sweet spot: $k = 5$ or $k = 10$

Why $k = 10$? A Visual Argument



Empirically, $k = 5$ or $k = 10$ gives the best bias-variance balance

CV for Classification

Same idea, but use **error rate** instead of MSE:

$$\mathrm{CV}_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{Err}_i \qquad \text{where } \mathrm{Err}_i = \frac{1}{n_i}\sum_{j \in C_i} \mathbf{1}(y_j \neq \hat{y}_j)$$

Example: Logistic on Default

Method	CV Error
Validation set	2.7%
LOOCV	2.6%
10-fold CV	2.6%

All give similar results here

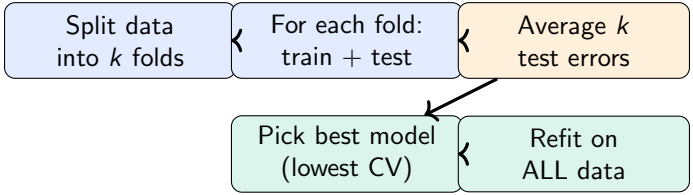
CV for model selection:

Which polynomial degree for logistic?

1. For each $d = 1, \ldots, 5$:

   Compute 10-fold CV error
2. Pick d with lowest CV error
3. Refit on full data with that d

CV in Practice: The Complete Workflow



Important: After selecting the best model via CV, **refit** it on the entire dataset to get the final model!

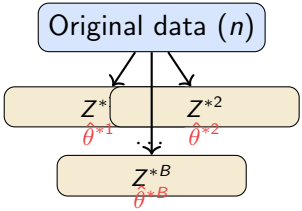
The Bootstrap: A Completely Different Idea

Problem:
How variable is our estimate $\hat{\theta}$?

"If I collected data again, how different would $\hat{\theta}$ be?"

Ideal (impossible):
Collect 1000 new datasets from the population, compute $\hat{\theta}$ each time

Bootstrap (practical):
Sample **with replacement** from our dataset, pretend each resample is a new dataset



Bootstrap: How It Works

Sampling with replacement:

Original data: $\{1, 2, 3, 4, 5\}$

Bootstrap sample 1: $\{2, 1, 4, 2, 5\}$
Bootstrap sample 2: $\{3, 3, 1, 5, 4\}$
Bootstrap sample 3: $\{5, 1, 1, 3, 2\}$

Same size n, but with **repeats**
Some observations appear multiple times; others are left out

On average:
Each bootstrap sample contains $\approx 63.2\%$ of unique observations

$$P(\text{obs } j \text{ not in sample}) = \left(1 - \frac{1}{n}\right)^n \approx e^{-1} \approx$$

So $\approx 36.8\%$ are **out-of-bag** (OOB)

This is useful! (more in Week 6)

Bootstrap: The Algorithm

To estimate $\mathrm{SE}(\hat{\theta})$:

1. Repeat B times ($B = 1000$ or more):
   a. Draw a sample of size n **with replacement** from data
   b. Compute $\hat{\theta}^{*b}$ on this bootstrap sample
2. Estimate the standard error:

$$\mathrm{SE}_B(\hat{\theta}) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\theta}^{*b} - \bar{\hat{\theta}}^{*}\right)^2}$$

where $\bar{\hat{\theta}}^{*} = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}^{*b}$

The variability across bootstrap samples mimics the variability across real datasets!

Bootstrap: Worked Example — Portfolio Allocation

Two assets: X and Y
Invest fraction α in X, $(1-\alpha)$ in Y

Optimal allocation:

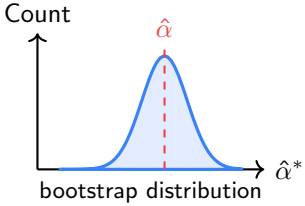
$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

minimizes $\mathrm{Var}(\alpha X + (1-\alpha)Y)$

α depends on $\sigma_X^2, \sigma_Y^2, \sigma_{XY}$
which we estimate from data
How variable is $\hat{\alpha}$?

Bootstrap results:
$B = 1000$ bootstrap samples

$\hat{\alpha} = 0.576$
$\mathrm{SE}_B(\hat{\alpha}) = 0.089$



Bootstrap for Regression Coefficients

Goal: Estimate $\text{SE}(\hat{\beta}_0)$ and $\text{SE}(\hat{\beta}_1)$ for SLR

Formula approach (Week 1):

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\sigma^2}{\sum(x_i - \bar{x})^2}}$$

Assumes: linear model, constant σ^2, independent errors

Bootstrap approach:
No assumptions needed!

Auto data results:

	Formula SE	Bootstrap SE
$\hat{\beta}_0$	0.717	0.860
$\hat{\beta}_1$	0.006	0.007

Bootstrap SE is **larger**!
Why? The formula assumes linearity, but the true relationship is non-linear $\Rightarrow$ formula SEs are too optimistic

When Bootstrap SEs Differ from Formula SEs

If assumptions hold:
Formula SE $\approx$ Bootstrap SE
Both are valid

If assumptions are violated:
Bootstrap SE $>$ Formula SE (usually)

Trust the bootstrap!
It accounts for model misspecification

Common violations:

- Non-linearity
- Heteroscedasticity
- Correlated errors
- Outliers

The bootstrap "sees" these problems; the formula doesn't

$\boxed{\text{Bootstrap} = \text{model-free uncertainty}}$

Think About It: What Does "Test Error" Really Mean?

Test MSE:

$$\text{Err} = E\left[(Y - \hat{f}(X))^2\right]$$

Expectation over **new** (X, Y) pairs drawn from the same population

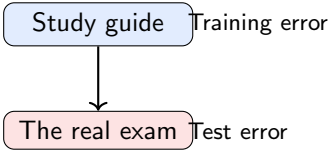
Training MSE:

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$$

Computed on the **same data** used to build $\hat{f}$

Always $\leq$ test MSE (often much less)

Analogy:

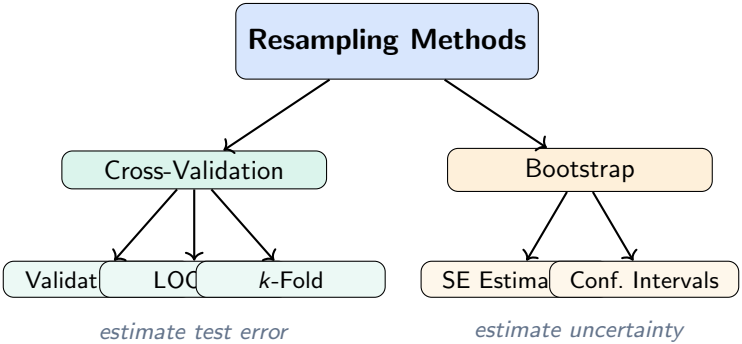


You need the real exam!

Memorizing answers $\neq$ understanding

Overfitting = memorization

The Resampling Toolbox: Big Picture



Validation Set: Step-by-Step with Numbers

Auto data: $n = 392$
$Y = \text{mpg}$, $X = \text{horsepower}$

Step 1: Split randomly
Train: obs $\{3, 7, 1, 9, 12, \ldots\}$ $(n = 196)$
Test: obs $\{2, 4, 5, 6, 8, \ldots\}$ $(n = 196)$

Step 2: Fit on train only
$\hat{Y} = 40.3 - 0.162X$

Step 3: Predict on test
$\hat{y}_2 = 40.3 - 0.162 \times 130 = 19.24$
Actual: $y_2 = 15.0$, $\text{error}^2 = 17.98$

Step 4: Average all squared errors

Obs	y_i	$\hat{y}_i$	$(y - \hat{y})^2$
2	15.0	19.2	17.98
4	16.0	15.8	0.04
5	17.0	18.5	2.25
$\vdots$			$\vdots$

$$\text{Val MSE} = \frac{1}{196} \sum_{i \in \text{test}} (y_i - \hat{y}_i)^2$$

≈ 23.6 (depends on split!)

Validation Set: Three Splits, Three Answers

Degree	Seed 1	Seed 2	Seed 3
1 (linear)	23.62	25.73	22.41
2 (quadratic)	18.76	20.43	17.95
3 (cubic)	18.80	20.38	18.12
5	19.03	21.12	17.84
10	19.49	22.58	18.53

- MSE varies by **2–3 units** depending on the split
- Quadratic is always better than linear — consistent
- But exact MSE values are **unreliable** — inconsistent
- "Best" degree varies: sometimes 2, sometimes 5

Lesson: We need a method that averages over many splits

LOOCV: What Happens Inside

Iteration $i = 1$:

1	2	3	4	5	6	7	8	9	10

Train on $\{2, 3, \ldots, n\}$, test on $\{1\}$
$\text{MSE}_1 = (y_1 - \hat{y}_1^{(-1)})^2$

Iteration $i = 2$:

1	2	3	4	5	6	7	8	9	10

Train on $\{1, 3, \ldots, n\}$, test on $\{2\}$
$\text{MSE}_2 = (y_2 - \hat{y}_2^{(-2)})^2$

$\vdots$ (repeat for all n observations)

$$\text{CV}_{(n)} = \frac{1}{n}(\text{MSE}_1 + \text{MSE}_2 + \cdots + \text{MSE}_n)$$

Every observation gets a turn as the test point!

LOOCV: Numerical Computation (Mini Example)

Toy data: $n = 5$, simple linear regression $Y = \beta_0 + \beta_1 X$

x_i	y_i
1	2.1
2	3.9
3	5.0
4	6.8
5	8.2

Leave out obs 3:

Train on $\{(1, 2.1), (2, 3.9), (4, 6.8), (5, 8.2)\}$

Fit: $\hat{Y} = 0.43 + 1.54X$

Predict: $\hat{y}_3 = 0.43 + 1.54(3) = 5.05$

$\text{MSE}_3 = (5.0 - 5.05)^2 = 0.0025$

All iterations:

Leave out	$\hat{y}_i^{(-i)}$	e_i	e_i^2
1	1.88	0.22	0.048
2	3.88	0.02	0.000
3	5.05	-0.05	0.003
4	6.72	0.08	0.006
5	8.22	0.02	0.000

The Shortcut Formula: Leverage-Adjusted Residuals

Regular LOOCV: Refit model n times $\Rightarrow$ expensive
Shortcut: Fit model **once**, adjust residuals by leverage

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{1 - h_i}\right)^2$$

Why does this work?
High-leverage point: h_i close to 1
$\Rightarrow$ Point strongly influences its own fit
$\Rightarrow$ Residual e_i is artificially small
$\Rightarrow$ Dividing by $(1 - h_i)$ inflates it back

Numerical verification:

i	e_i	h_i	$\frac{e_i}{1-h_i}$
1	0.18	0.40	0.30
2	0.02	0.20	0.03
3	−0.04	0.13	−0.05
4	0.06	0.20	0.08
5	−0.01	0.40	−0.02

Edge obs (1,5): largest h_i, largest correction

k-Fold CV: Worked Example with $k = 5$

Auto data: $n = 392$, quadratic model, $k = 5$
Each fold has ≈ 78 observations

Fold	Train n	Test n	MSE_i
1	314	78	18.95
2	314	78	19.82
3	314	78	19.41
4	314	78	18.73
5	314	78	19.34

$$\text{CV}_{(5)} = \frac{18.95 + 19.82 + 19.41 + 18.73 + 19.34}{5} = 19.25$$

Very close to the LOOCV estimate of 19.25 — but 78× faster!

Stratified k-Fold CV: For Classification

Problem: With imbalanced classes, a random fold might get **zero** positive examples!

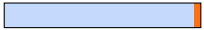
Default data: 3.3% default rate
In 10-fold: each fold has $\sim$ 1000 obs
Expected defaults per fold: $\sim$ 33

Stratified CV:
Each fold has the **same proportion** of each class as the full dataset
$\Rightarrow$ More stable error estimates

Regular:

might be unbalanced

Stratified:



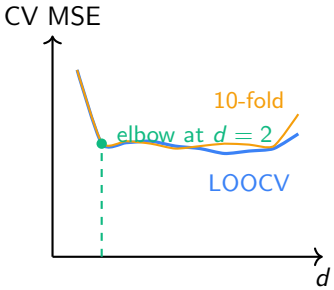
same 3.3% in every fold

R: `caret::createFolds(y, k=10)` automatically stratifies

CV for Model Selection: Polynomial Degree

Goal: Choose d in $\text{mpg} \sim \text{poly}(\text{hp}, d)$

d	LOOCV	10-fold
1	24.23	24.27
2	19.25	19.27
3	19.33	19.35
4	19.42	19.29
5	19.03	19.03
7	18.83	19.12
10	19.49	20.96



Both methods agree:
$d = 2$ is the sweet spot
Higher degrees add complexity without reducing error

CV for Model Selection: Comparing Entire Models

Which model predicts mpg best?

Model	10-fold CV MSE	# predictors
mpg $\sim$ horsepower	24.27	1
mpg $\sim$ poly(hp, 2)	19.27	2
mpg $\sim$ weight	18.95	1
mpg $\sim$ weight + year	14.82	2
mpg $\sim$ weight + year + origin	12.55	3
mpg $\sim$. − name	11.58	7

- Weight alone beats poly(hp, 2)!
- Adding year provides a large improvement
- Full model is best but uses 7 predictors
- **Weight + year + origin**: good trade-off

CV for Classification: Default Dataset

Logistic regression:
default $\sim$ balance

10-fold CV error rate:

Fold	Error rate
1	2.3%
2	2.8%
3	2.5%
$\vdots$	$\vdots$
10	2.7%
Average	**2.6%**

But what about sensitivity?

10-fold CV can also report:

- Average sensitivity per fold
- Average AUC per fold
- Average F1 per fold

Accuracy alone is misleading!
(Recall: predicting "No" always gives 96.7% accuracy)

R: Use `caret::train()` for classification metrics per fold

The "One Standard Error Rule"

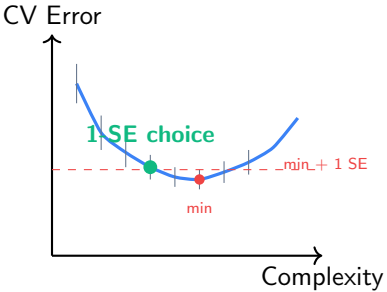
Problem:
The model with the *lowest* CV error may be too complex

One-SE Rule:
Choose the **simplest model** whose CV error is within one standard error of the minimum

Why?

- Simpler models are more interpretable
- The "best" model might be winning by noise
- Conservative: prefer parsimony

Used heavily in Lasso/Ridge (Week 4)



Repeated k-Fold CV

Problem: Even 10-fold CV has some randomness (from the random split into folds)

Solution: Repeat the entire k-fold procedure R times

1. Run 10-fold CV with seed 1 $\Rightarrow$ CV_1
2. Run 10-fold CV with seed 2 $\Rightarrow$ CV_2

   $\vdots$

R. Run 10-fold CV with seed R $\Rightarrow$ CV_R

Final estimate: $\frac{1}{R}\sum_{r=1}^{R} \text{CV}_r$

How many repeats?

R	SE reduction
1	baseline
5	$\div\sqrt{5} \approx 2.2\times$
10	$\div\sqrt{10} \approx 3.2\times$

Typical: $R = 5$ or $R = 10$

R: `caret::trainControl(`
`method="repeatedcv",`
`number=10, repeats=5)`

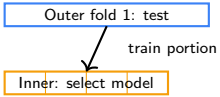
Nested CV: The Gold Standard

Problem:
If we use CV to **select** a model AND **estimate** its error, the estimate is biased (optimistic)

Solution: Nested CV

- **Outer loop:** 10-fold CV estimates test error
- **Inner loop:** 5-fold CV selects the model

The inner loop tunes, the outer loop evaluates



Repeat for all outer folds

Computationally expensive but unbiased

Bootstrap: The Key Insight

Why does resampling with replacement mimic new datasets?

The plug-in principle:

We don't know the population distribution F

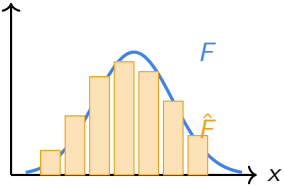
But our sample $\hat{F}$ is our best estimate of F

Drawing from $\hat{F}$ with replacement $\approx$ drawing from F

As $n \to \infty$:

$\hat{F} \to F$, so bootstrap $\to$ exact

For finite n: approximate, but usually very good



The histogram approximates the true density

Bootstrap: Which Observations Get Selected?

Each bootstrap sample: draw n times with replacement

Original: $\{1, 2, 3, 4, 5, 6, 7, 8\}$
Sample 1: $\{2, 5, 5, 1, 7, 3, 5, 8\}$
$\Rightarrow$ Obs 5 appears 3 times!
$\Rightarrow$ Obs 4, 6 are missing

Sample 2: $\{3, 1, 1, 6, 4, 3, 8, 2\}$
$\Rightarrow$ Obs 1, 3 appear twice
$\Rightarrow$ Obs 5, 7 are missing

Probability of inclusion:
$P(\text{obs } j \text{ included}) = 1 - \left(1 - \frac{1}{n}\right)^n$

n	$P(\text{included})$
5	67.2%
10	65.1%
100	63.4%
∞	$1 - e^{-1} = 63.2\%$

$\approx 36.8\%$ are **out-of-bag (OOB)**

Bootstrap SE: Step-by-Step Example

Estimate the SE of the sample mean $\bar{X}$
Original data: $\{3, 7, 2, 5, 8\}$
$\bar{X} = 5.0$

$B = 4$ bootstrap samples:

Sample	$\bar{X}^{*b}$
$\{7, 3, 3, 5, 7\}$	5.0
$\{8, 2, 7, 2, 5\}$	4.8
$\{5, 5, 8, 3, 2\}$	4.6
$\{3, 8, 7, 7, 5\}$	6.0

$$\bar{\hat{X}}^* = (5.0 + 4.8 + 4.6 + 6.0)/4 = 5.1$$

$$\text{SE}_B = \sqrt{\frac{1}{3}\sum_{b=1}^{4}(\bar{X}^{*b} - 5.1)^2}$$

$$= \sqrt{\frac{(0.01+0.09+0.25+0.81)}{3}}$$

$$= \sqrt{0.387} = 0.622$$

Formula SE: $s/\sqrt{n} = 2.55/\sqrt{5} = 1.14$

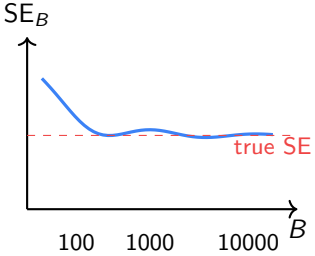
(With only $B = 4$, bootstrap is noisy. Use $B \geq 1000$!)

How Many Bootstrap Samples B?

Rule of thumb:

Goal	B
Standard errors	$B \geq 1{,}000$
Confidence intervals	$B \geq 5{,}000$
Hypothesis testing	$B \geq 10{,}000$

More B = more precise SE estimate
But diminishing returns past $B \approx 10{,}000$



Stabilizes around $B = 1000$

Bootstrap vs. Formula: When They Disagree

Auto data: SLR (mpg $\sim$ hp)

	Formula	Bootstrap
SE($\hat{\beta}_0$)	0.717	0.860
SE($\hat{\beta}_1$)	0.006	0.007

Bootstrap SEs are **larger**!

Why?

Formula assumes $Y = \beta_0 + \beta_1 X + \varepsilon$

But true relationship is non-linear

$\Rightarrow$ Formula SEs are too optimistic

Auto data: Quadratic (mpg $\sim$ hp + hp^2)

	Formula	Bootstrap
SE($\hat{\beta}_0$)	1.800	1.870
SE($\hat{\beta}_1$)	0.031	0.030
SE($\hat{\beta}_2$)	0.0001	0.0001

Now they **agree**!

Quadratic model is correct

$\Rightarrow$ formula assumptions hold

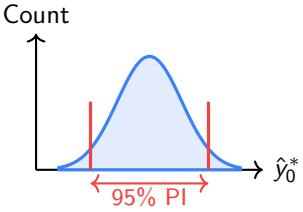
$\Rightarrow$ both methods give same answer

Bootstrap for Prediction Intervals

Beyond standard errors:
Bootstrap can estimate **any** quantity's distribution

Prediction at $x_0 = 100$ hp:

1. For each bootstrap sample b:
   a. Fit model on sample b
   b. Predict $\hat{y}_0^{*b}$ at $x_0 = 100$
   c. Add noise: $y_0^{*b} = \hat{y}_0^{*b} + \varepsilon^{*b}$
2. Use percentiles of $\{y_0^{*1}, \ldots, y_0^{*B}\}$ as prediction interval



Much wider than the confidence interval!

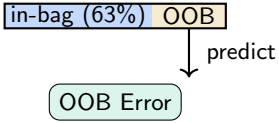
Out-of-Bag Error: Bootstrap Meets CV

Key observation: Each bootstrap sample leaves out $\sim 36.8\%$ of observations

OOB error estimation:

1. For bootstrap sample b, predict the $\sim 36.8\%$ left-out observations
2. Average these predictions across all bootstrap samples
3. This gives an estimate of test error!

Advantage:
No separate validation set needed — it's "free" from the bootstrap



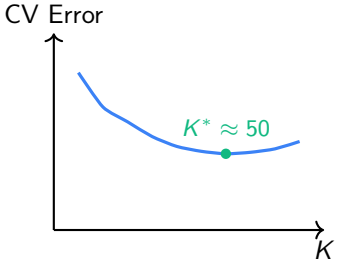
Used in Random Forests (Week 6) as the default error estimate!

CV for Tuning Hyperparameters: KNN Example

Question: What is the best K for KNN on the Default data?

10-fold CV results:

K	CV Error Rate
1	3.8%
3	3.2%
5	2.9%
10	2.7%
20	2.6%
50	2.5%
100	2.6%



Workflow:

1. CV for each K
2. Pick K^* with lowest CV error
3. Refit KNN with K^* on all data

Data Leakage: The Silent Killer

Wrong:

1. Standardize ALL data
2. Then split into folds
3. Run CV

The test fold "knows" the mean and SD of the training data
$\Rightarrow$ CV error is **optimistic**
$\Rightarrow$ Real test error is **worse**

Right:

1. Split into folds
2. For each fold:
   a. Compute mean/SD on **train only**
   b. Standardize train with those stats
   c. Apply **same** stats to test fold

Test fold is truly unseen!

Rule: Any computation on data must happen **inside** each CV fold,
not before splitting!

Bootstrap Confidence Intervals: Three Methods

Method	Formula	Assumes Normal?
Normal	$\hat{\theta} \pm z_{\alpha/2} \cdot \text{SE}_B$	Yes
Percentile	$[\hat{\theta}^*_{\alpha/2},\ \hat{\theta}^*_{1-\alpha/2}]$	No
BCa (bias-corrected)	Adjusted percentiles	No

Portfolio example ($\hat{\alpha} = 0.576$):

Method	95% CI
Normal	[0.40, 0.75]
Percentile	[0.39, 0.74]
BCa	[0.41, 0.76]

All three are similar here

When they differ:

If $\hat{\theta}^*$ distribution is **skewed**, percentile and BCa are better

R: `boot.ci(boot.out, type=c("norm","perc","bca"))`

Parametric vs. Non-Parametric Bootstrap

Non-parametric (standard):
Resample (x_i, y_i) pairs with replacement
No model assumptions

What we've been doing!

Use when:

- Model may be wrong
- Want robust SEs
- General-purpose

Parametric:

1. Fit model: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$
2. Generate new $Y^* = \hat{Y} + \varepsilon^*$ where $\varepsilon^* \sim N(0, \hat{\sigma}^2)$
3. Refit on (X, Y^*)

Use when:

- Model is trusted
- Want more efficiency
- Especially with small n

What Can Go Wrong with the Bootstrap?

1. **Small sample size**
   - $n < 20$: bootstrap distribution may be poor
   - Not enough "diversity" in resamples
2. **Dependent data**
   - Time series: observations are correlated
   - Standard bootstrap breaks independence assumption
   - Use **block bootstrap** instead
3. **Extreme values / heavy tails**
   - Bootstrap may miss rare events
   - Large outliers dominate some resamples
4. **Not enough B**
   - $B = 50$ is not enough! Use $B \geq 1000$

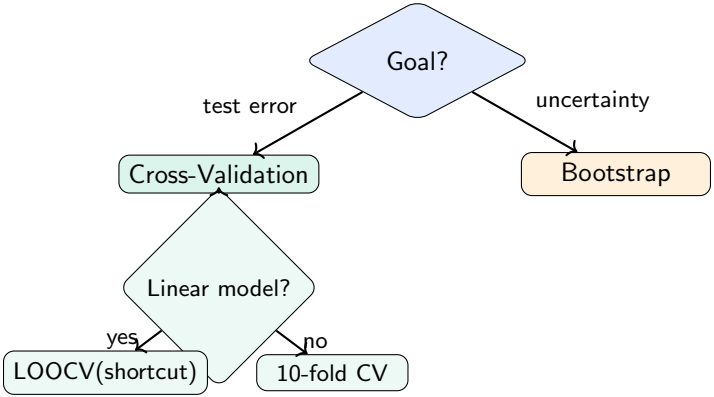
Computational Cost Comparison

Method	Model fits	Auto ($n = 392$)	Default ($n = 10$k)
Validation set	1	instant	instant
5-fold CV	5	instant	fast
10-fold CV	10	instant	fast
LOOCV	n	392 fits	10,000 fits
LOOCV (shortcut)	1	instant	instant
Bootstrap (B=1000)	1,000	fast	moderate

Key points:

- The LOOCV shortcut only works for **linear** models
- For complex models (trees, neural nets): LOOCV is prohibitive
- 10-fold CV: always practical — the default choice

When to Use Which Method: Decision Guide



Real-World Example: Choosing a Spam Filter

Scenario: Compare 4 classifiers for spam detection

Model	10-fold CV Error	CV Sensitivity	CV AUC
Logistic	7.2%	82%	0.94
LDA	8.1%	78%	0.91
QDA	5.5%	88%	0.96
KNN ($K = 5$)	6.8%	85%	0.93

Steps taken:

1. 10-fold CV to estimate error for each method
2. Report sensitivity because missing spam is costly
3. Choose QDA: best on all three metrics
4. Refit QDA on the full dataset for deployment

Real-World Example: Predicting House Prices

Boston data: Which polynomial degree for medv $\sim$ lstat?

Degree	10-fold CV MSE
1	38.5
2	30.1
3	28.8
4	28.2
5	28.3
10	30.5

Degree 4 is minimum
but 1-SE rule $\Rightarrow$ degree 2 or 3

Bootstrap SE for degree 2:

$\hat{\beta}_0 = 42.9 \pm 0.97$
$\hat{\beta}_1 = -2.33 \pm 0.12$
$\hat{\beta}_2 = 0.044 \pm 0.004$

All coefficients significant

Final model:

$$\widehat{\text{medv}} = 42.9 - 2.33 \cdot \text{lstat} + 0.044 \cdot \text{lstat}^2$$

CV for Logistic Regression: Degree of Polynomial

Default data: default $\sim$ poly(balance, d)

Degree	10-fold CV Error
1	2.67%
2	2.61%
3	2.62%
4	2.62%

Minimal improvement beyond $d = 1$
Logistic is already very flexible
for this dataset

Interpretation:
The relationship between balance
and log-odds of default is
approximately linear

Adding polynomial terms does not help

$\Rightarrow$ Stick with the simple logistic model

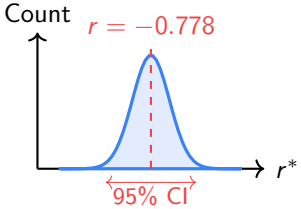
Bootstrap for Correlation Coefficient

How variable is r_{XY}?

Auto data: $r(\text{horsepower, mpg}) = -0.778$

$B = 1000$ bootstrap:
$\text{SE}_B(r) = 0.019$
95% CI: $[-0.815, \ -0.740]$

Very precise!
The negative correlation is extremely robust



No formula needed!
Bootstrap handles any statistic

Bootstrap for Median: No Formula Exists!

The bootstrap shines when no formula is available

Boston data:
Median home value: \$21,200

SE of the median?
No closed-form formula!

Bootstrap ($B = 1000$):
$\text{SE}_B(\text{median}) = \$0.38\text{k}$
95% CI: [\$20.5k, \$22.0k]

Compare with mean:
Mean home value: \$22,533
Formula SE: \$0.41k
Bootstrap SE: \$0.41k

For the mean: formula and bootstrap agree (as expected)
For the median: **only bootstrap works**

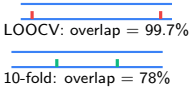
Bootstrap is universal!

Think About It: Why $k = n$ Has High Variance

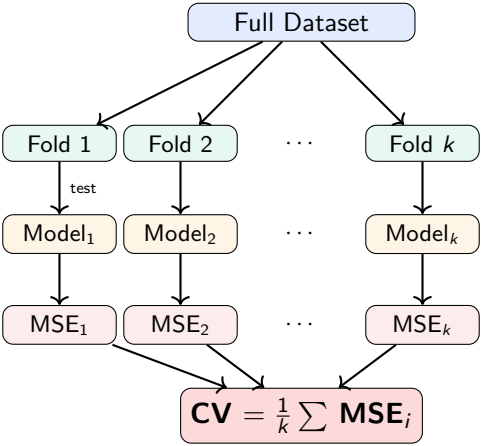
LOOCV: average n estimates
Each trained on $n - 1$ observations
$\Rightarrow$ Training sets differ by only 1 point!
$\Rightarrow$ $\text{MSE}_1, \text{MSE}_2, \ldots, \text{MSE}_n$ are highly **correlated**

Recall:
$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} + \frac{n-1}{n}\rho\sigma^2$
When ρ is large, variance of the average is large!

$k = 10$**:** average 10 estimates
Each trained on 90% of data
Training sets overlap less
$\Rightarrow$ $\text{MSE}_1, \ldots, \text{MSE}_{10}$ are less correlated
$\Rightarrow$ Lower variance of the average



The Cross-Validation Cycle: A Visual Summary



Worked Example: 5-Fold CV by Hand

Data: 10 observations, 5-fold CV

Fold	Train	Test	MSE_i
1	obs 3–10	obs 1–2	2.1
2	obs 1,2,5–10	obs 3–4	1.8
3	obs 1–4,7–10	obs 5–6	2.5
4	obs 1–6,9–10	obs 7–8	1.9
5	obs 1–8	obs 9–10	2.3

$$\text{CV}_{(5)} = \frac{2.1 + 1.8 + 2.5 + 1.9 + 2.3}{5}$$

$$= \frac{10.6}{5} = 2.12$$

Each observation appears in the test set **exactly once**

Every observation contributes to the error estimate

The LOOCV Shortcut: Why It Works

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{1-h_i}\right)^2$$

Intuition:

$e_i = y_i - \hat{y}_i$ = ordinary residual

h_i = leverage of observation i

High leverage $\Rightarrow$ observation strongly influences its own fit

$\Rightarrow$ Ordinary residual **underestimates** how much this point deviates

$\Rightarrow$ Dividing by $(1 - h_i)$ **corrects** this

Numerical check:

i	e_i	h_i	$\frac{e_i}{1-h_i}$
1	0.5	0.05	0.53
2	-0.8	0.02	-0.82
3	1.2	0.15	1.41
4	-0.3	0.30	-0.43

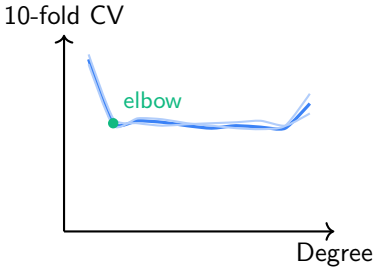
Obs 4: high leverage $\Rightarrow$ big correction

CV for Choosing the Right Polynomial Degree

Auto data: 10-fold CV

Degree	CV MSE
1 (linear)	24.27
2 (quadratic)	19.27
3 (cubic)	19.35
4	19.29
5	19.03

Conclusion: Quadratic is sufficient
Higher degrees add complexity
without meaningful improvement



9 different random splits
of 10-fold CV all agree!

Bootstrap Confidence Intervals

Two common approaches:

Normal bootstrap CI:

$$\hat{\theta} \pm z_{\alpha/2} \cdot \mathrm{SE}_B(\hat{\theta})$$

Assumes $\hat{\theta}$ is approximately normal
Simple but can be inaccurate

Example:
$\hat{\alpha} \pm 1.96 \times 0.089$
$= [0.40,\ 0.75]$

Percentile bootstrap CI:
Use the $\alpha/2$ and $1 - \alpha/2$ quantiles of the bootstrap distribution directly

No normality assumption!

Example:
2.5th percentile: 0.39
97.5th percentile: 0.74
$\Rightarrow$ 95% CI: [0.39, 0.74]

Common Mistakes with CV

1. **Data leakage:** Preprocessing *before* splitting
   - Wrong: Standardize all data, then split into folds
   - Right: Standardize training folds only, apply to test fold
2. **Using CV error as the final test error**
   - CV is for model **selection**, not final assessment
   - Ideally, hold out a separate test set that is never touched during CV
3. **Forgetting to refit after CV**
   - After choosing the best model via CV, refit on **all** data
4. **Too few folds with small n**
   - $k = 10$ with $n = 30 \Rightarrow$ only 3 test obs per fold $\Rightarrow$ noisy

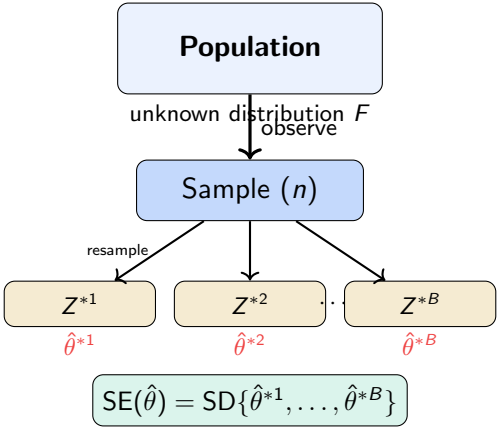
Bootstrap vs. Cross-Validation: Different Questions

	Cross-Validation	Bootstrap
Purpose	Estimate **test error**	Estimate **uncertainty**
Question	How well does model predict?	How variable is $\hat{\theta}$?
Resampling	Split into folds (no replacement)	Sample with replacement
Output	CV error (MSE or error rate)	SE($\hat{\theta}$) or CI
Use for	Model selection, tuning	Confidence intervals, SEs

Both are resampling methods, but they answer different questions!

CV: "Which model is best?" *Bootstrap: "How sure are we?"*

The Bootstrap in Pictures



Quick Quiz: Test Yourself!

1. LOOCV trains on how many observations each time?

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   $n - 1$ observations (leaves out exactly 1)

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   $9/10 = 90\%$

3. Why does LOOCV have higher variance than 10-fold CV?

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   Training sets overlap almost completely $\Rightarrow$ estimates are highly correlated

4. What does the bootstrap sample **with** or **without** replacement?

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   $9/10 = 90\%$

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4. What does the bootstrap sample **with** or **without** replacement?

   With replacement — that's the key idea!

Quick Quiz: More Questions

5. What fraction of observations is NOT in a bootstrap sample (on average)?

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   $\approx$ 36.8% (out-of-bag observations)

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   Yes! Use misclassification rate instead of MSE

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   Quadratic — lower CV error, big improvement over linear

8. After selecting a model via CV, should you report the CV error as test error?

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8. After selecting a model via CV, should you report the CV error as test error?

   Ideally no — CV was used for selection, use a held-out test set for final assessment

R Functions: Quick Reference

Cross-Validation	
`cv.glm(data, glmfit)`	LOOCV (default)
`cv.glm(data, glmfit, K=10)`	10-fold CV
`cv.glm()$delta[1]`	Raw CV error
`cv.glm()$delta[2]`	Bias-corrected CV error

Bootstrap	
`boot(data, statistic, R=1000)`	Run R bootstrap replicates
`boot.out$t0`	Original estimate
`boot.out$t`	Matrix of bootstrap estimates
`boot.ci(boot.out)`	Bootstrap confidence intervals

Both from the `boot` package: `library(boot)`

Key Formulas: Cheat Sheet

Cross-Validation:

$$\mathrm{CV}_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{MSE}_i$$

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{1-h_i}\right)^2$$

Classification:

$$\mathrm{CV}_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{Err}_i$$

Bootstrap:

$$\mathrm{SE}_B(\hat{\theta}) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}(\hat{\theta}^{*b} - \bar{\hat{\theta}}^{*})^2}$$

Out-of-bag probability:

$$P(\text{not selected}) = \left(1 - \frac{1}{n}\right)^n \approx e^{-1}$$

Normal bootstrap CI:

$$\hat{\theta} \pm z_{\alpha/2} \cdot \mathrm{SE}_B$$

Summary: Chapter 5

Cross-Validation:

- Estimates test error without a test set
- Validation set: simple but high variance
- LOOCV: low bias, high variance, expensive
- k-fold: best bias-variance balance at $k = 5$–10
- Use for model selection and tuning

Bootstrap:

- Estimates uncertainty of any statistic
- Sample with replacement, B times
- No distributional assumptions
- Provides SEs and CIs
- More honest than formula-based SEs when model assumptions fail

Next week: Regularization — what to do when you have too many predictors

Implementing CV Manually in R

```
set.seed(1)
k <- 10
n <- nrow(Auto)
folds <- sample(rep(1:k, length=n))
cv_errors <- rep(0, k)

for (i in 1:k) {
  test_idx <- which(folds == i)
  fit <- lm(mpg ~ poly(hp,2),
        data=Auto[-test_idx,])
  pred <- predict(fit, Auto[test_idx,])
  cv_errors[i] <- mean(
    (Auto$mpg[test_idx] - pred)^2)
}
mean(cv_errors) # CV estimate
```

Key steps:

1. Create fold assignments
2. Loop over folds
3. Train on $-$fold, test on fold
4. Average MSEs

Understanding the mechanics helps you debug and customize!

Or use:

`cv.glm(data, fit, K=10)`

Implementing Bootstrap Manually in R

```
B <- 1000
n <- nrow(Auto)
beta1_boot <- rep(0, B)

for (b in 1:B) {
  idx <- sample(n, replace=TRUE)
  fit <- lm(mpg ~ horsepower,
        data=Auto[idx,])
  beta1_boot[b] <- coef(fit)[2]
}

sd(beta1_boot) # bootstrap SE
quantile(beta1_boot, c(.025,.975))
```

Key: `replace = TRUE`
This is what makes it bootstrap!

Results:
$\text{SE}_B(\hat{\beta}_1) \approx 0.007$
95% CI: $[-0.172, -0.144]$

Or use:
`boot(data, statistic, R=1000)`

The `boot()` Function: Anatomy

Three ingredients:

1. **Data:** The original dataset
2. **Statistic:** A function that takes (data, index) and returns $\hat{\theta}$
3. **R:** Number of bootstrap replicates

The statistic function: `my.fn <-`

```
function(data, index) {
  d <- data[index, ]
  fit <- lm(mpg ~ hp, data=d)
  return(coef(fit))
}
```

`index` = vector of row indices

`boot()` generates these with replacement

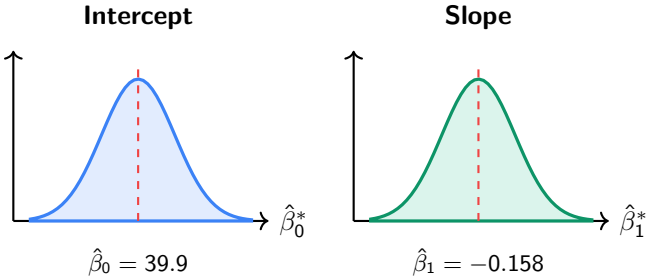
Running it: `library(boot)`

```
result <- boot(Auto, my.fn, R=1000)
result$t0 # original estimate
result$t # B x p matrix
boot.ci(result) # CIs
```

`t0` = $\hat{\theta}$ from original data

`t` = all B bootstrap estimates

Visualizing Bootstrap Distributions



- Both distributions are approximately normal $\Rightarrow$ normal CI is fine
- Width of histogram $\propto$ SE
- The intercept has wider spread (larger SE) than the slope

CV for Choosing Between Methods (Not Just Tuning)

Question: Linear regression or KNN for predicting mpg?

10-fold CV:

Method	CV MSE
lm(mpg $\sim$ hp)	24.3
lm(mpg $\sim$ poly(hp,2))	19.3
KNN $K = 5$	20.1
KNN $K = 10$	20.8
KNN $K = 20$	22.5
lm(mpg $\sim$ wt+yr+orig)	12.6

Conclusion:

Linear regression with the right predictors beats KNN!

Why?

- The relationship is approximately linear
- KNN suffers from curse of dimensionality
- Linear model uses structure effectively

CV lets us compare apples to oranges fairly!

Double Use of Data: A Subtle Error

Common mistake in practice:

Wrong:

1. Look at all data to choose features
2. Then run CV on those features
3. Report CV error

The feature selection "saw" the test data!
$\Rightarrow$ CV error is too optimistic

Right:

1. Split into folds
2. **Inside each fold:**
   1. Select features on train
   2. Fit model on train
   3. Test on held-out fold

Feature selection is part of the model-building pipeline — it must be inside CV!

Golden rule: Everything that uses data to make decisions must be **inside** the CV loop

LOOCV for Classification: Worked Example

Default data: $n = 10{,}000$, logistic regression

Iteration $i = 1$:
Train on obs $\{2, 3, \ldots, 10000\}$
Predict obs 1: $\hat{p}_1 = 0.02$
Actual: default = No
$\Rightarrow \text{Err}_1 = \mathbf{1}(\text{No} \neq \text{No}) = 0$ ✓

Iteration $i = 47$:
Train without obs 47
Predict: $\hat{p}_{47} = 0.62$
Actual: default = No
$\Rightarrow \text{Err}_{47} = \mathbf{1}(\text{Yes} \neq \text{No}) = 1$ ×

After all n iterations:

$$\text{CV}_{(n)} = \frac{1}{10000} \sum_{i=1}^{10000} \text{Err}_i$$

$= \frac{264}{10000} = 2.64\%$

264 observations were misclassified when left out

For logistic, LOOCV shortcut doesn't apply — must refit n times (slow!)
Better: use 10-fold CV

Sample Size and CV Stability

How stable is 10-fold CV?

Run 10-fold CV 100 times with different random folds:

n	Mean CV	SD of CV
50	22.3	3.1
100	20.8	1.8
200	19.5	0.9
392	19.3	0.4

Key insight:

Larger $n \Rightarrow$ more stable CV estimates

With $n = 50$: CV varies a lot (SD $= 3.1$)
With $n = 392$: very stable (SD $= 0.4$)

Small datasets: consider LOOCV or repeated CV for more stable estimates

Bootstrap: Estimating SE of Any Statistic

The bootstrap works for ANY computable statistic!

Statistic	Formula SE exists?	Bootstrap works?
Mean $\bar{X}$	Yes ($s/\sqrt{n}$)	Yes
Regression $\hat{\beta}$	Yes (under assumptions)	Yes
Median	No simple formula	Yes
Correlation r	Complex (Fisher's z)	Yes
Ratio $\bar{X}/\bar{Y}$	No	Yes
90th percentile	No	Yes
Custom $\hat{\alpha}$	No	Yes

The bootstrap is universal — that's its superpower

Residual Bootstrap vs. Pairs Bootstrap

Pairs bootstrap (standard):
Resample (x_i, y_i) pairs together
$\{(x_3, y_3), (x_1, y_1), (x_3, y_3), \ldots\}$

- No model assumptions
- Handles heteroscedasticity
- May change leverage structure

Residual bootstrap:

1. Fit model: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$
2. Compute residuals: $e_i = y_i - \hat{y}_i$
3. Resample residuals: e^*
4. New data: $y_i^* = \hat{y}_i + e_i^*$
5. Refit on (x_i, y_i^*)

- Preserves X structure
- Assumes constant variance

In practice, pairs bootstrap is more common — safer and simpler

Connecting CV to Weeks 4–10

CV will be used in every remaining week:

Week	CV used for
4	Choose λ in Lasso/Ridge
5	Choose # knots in splines
6	Estimate RF test error
7	Tune boosting parameters
8	Choose cost C in SVM
9	Early stopping in neural nets
10	Choose # clusters

The bootstrap too:

- Week 6: Out-of-bag error in Random Forests
- Week 7: Bagging = bootstrap aggregation
- Confidence intervals everywhere

Today's methods are the foundation for everything ahead

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   Approximately yes, but it has some bias (slightly overestimates true test error)

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   Yes in principle, but each refit takes hours $\Rightarrow$ impractical. Use dropout instead.

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   Otherwise the test fold "leaks" into the selection step $\Rightarrow$ optimistic CV error

Exercise 1: Validation Set on Default (ISLR 5.4 Q5)

Data: `Default` Logistic: `default ~ income + balance`

1. Fit logistic on full data
2. Split into train/validation, fit on train, compute validation error
3. Repeat 3 times with different seeds — how much does error vary?
4. Add `student` — does validation error improve?

Key insight: Error varies $\pm 0.5\%$ across splits

Exercise 2: Bootstrap SEs for Logistic (ISLR 5.4 Q6)

Data: `Default`

1. Fit logistic: `default ~ income + balance`
2. Compute formula-based SEs from `summary()`
3. Write a `boot.fn()` function that returns coefficients
4. Use `boot()` with $B = 1000$ to get bootstrap SEs
5. Compare formula SEs vs. bootstrap SEs

Exercise 3: LOOCV on Auto (ISLR 5.4 Q5-6)

Data: `Auto`

`mpg ~ poly(horsepower, d)`

1. Compute LOOCV error for $d = 1, \ldots, 10$
2. Which degree minimizes LOOCV?
3. Repeat with 10-fold CV — same conclusion?
4. Bootstrap SE for the linear regression coefficients
5. Compare bootstrap SEs to `summary()` SEs

Hint: `cv.glm(Auto, glm.fit)$delta[1]` for LOOCV

Exercise 4: Bootstrap on Portfolio (ISLR 5.4 Q8)

Data: `Portfolio`

Estimate optimal α

1. Write a function to compute $\hat{\alpha}$ from the formula
2. Use `boot()` with $B = 1000$
3. Report $\hat{\alpha}$ and $\mathrm{SE}_B(\hat{\alpha})$
4. Construct a 95% confidence interval

Solution: Exercise 1 — Validation Set

Three different splits:

Seed	Validation Error
1	2.6%
42	2.8%
123	2.4%

Error varies by $\sim 0.4\%$

Adding `student`: no meaningful reduction

Key takeaway:

The validation set approach gives **variable** results

This is why k-fold CV is preferred — it averages over multiple splits

`student` adds nothing beyond `balance` (as we saw in Week 2)

Solution: Exercise 3 — LOOCV on Auto

LOOCV results:

Degree	LOOCV MSE
1	24.23
2	19.25
3	19.33
5	19.03
10	19.49

10-fold CV: Same conclusion — quadratic is the elbow

Bootstrap vs. Formula SEs:

	Formula	Bootstrap
$\hat{\beta}_0$	0.717	0.860
$\hat{\beta}_1$	0.006	0.007

Bootstrap SEs are larger
$\Rightarrow$ Formula underestimates uncertainty because true relationship is non-linear

Solution: Exercise 4 — Bootstrap on Portfolio

R code outline:

```
alpha.fn <- function(data, index) {
  X <- data$X[index]
  Y <- data$Y[index]
  (var(Y) - cov(X,Y)) /
  (var(X) + var(Y) - 2*cov(X,Y))
}

boot(Portfolio, alpha.fn, R=1000)
```

Results:

$\hat{\alpha} = 0.576$

$\text{SE}_B(\hat{\alpha}) = 0.089$

95% CI:

Normal: [0.40, 0.75]

Percentile: [0.39, 0.74]

Invest 58% in X, 42% in Y with substantial uncertainty

What's Next?

Week 2: Classification

↓

Week 3: Resampling

↓

Week 4: Lasso & Ridge

Homework: ISLR Chapter 5: Q5, 6, 7, 8
Lab: `Lab_Week03.R` — CV and Bootstrap on Auto, Default, Portfolio
Next week: Too many predictors? ⇒ Lasso & Ridge Regression!

Thank You

Questions?

Dr. Endri Raço — Machine Learning with R

`github.com/endri81/ISLRv2-solutions`